

Brandon T. Radzom

PH.D. · ASTRONOMY

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Education

Indiana University (IU) - Bloomington

Bloomington, IN 47405-7000

PH.D. ASTRONOMY

Aug. 2020 - June 2026

- Thesis: *Observational and Numerical Constraints on the Dynamical Origins of Hot Jupiters*
- Advisor: Prof. Songhu Wang

M.A. ASTRONOMY

Aug. 2020 - May 2023

- Minor: Scientific Computing

University of Wisconsin (UW) - Madison

Madison, WI 53706-1507

B.S. ASTRONOMY-PHYSICS, B.S. PHYSICS

Sep. 2016 - May 2020

- Minor: Computer Science
- Thesis: *Characterizing AGN Activity in the SSA22 Field*
- Advisor: Prof. Amy Barger

Professional Experience

- 2026 **Postdoctoral Scholar**, Center for Exoplanets and Habitable Worlds, Penn State
- 2026 **Visitor in Astrophysics**, Dept. of Astronomy, Caltech
- 2021-2026 **Graduate Research Assistant**, Dept. of Astronomy, IU-Bloomington
- 2020-2024 **Graduate Associate Instructor**, Dept. of Astronomy, IU-Bloomington
- 2018-2020 **Undergraduate Research Assistant**, Dept. of Astronomy, UW-Madison
- 2017-2020 **Undergraduate Research Assistant**, Dept. of Physics, UW-Madison

Publication List

FIRST-AUTHOR REFEREED

- Brandon T. Radzom**, Songhu Wang, Bonan Pu, Hareesh Gautham Bhaskar, and Malena Rice 2025. *Hot Jupiters' Isolation Is Not Unique to High-Eccentricity Tidal Migration*, resubmitted to ApJ, arXiv:2605.27362
- Brandon T. Radzom**, Jiayin Dong, Malena Rice, Xian-Yu Wang, Kyle Hixenbaugh, George Zhou, Chelsea X. Huang, Songhu Wang 2025. *Evidence for Primordial Alignment II: Insights from Stellar Obliquity Measurements for Hot Jupiters in Compact Multi-planet Systems*, AJ 169 189
- Brandon T. Radzom**, Jiayin Dong, Malena Rice, Xian-Yu Wang, Samuel W. Yee, Tyler R. Fairnington, Cristobal Petrovich, Songhu Wang 2024. *Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact Sub-Saturn Systems*, AJ 168 116
- Brandon T. Radzom**, Anthony J. Taylor, Amy J. Barger, Lennox L. Cowie 2022. *X-ray Sources in the Chandra Field SSA22*, ApJ 940 114

CONTRIBUTING-AUTHOR REFEREED

- Galen Bergsten, Ilaria Pascucci, Katherine Gonglewski, et al. (incl. **Brandon T. Radzom**) 2026. *Using Integrated Demographics to Constrain the Occurrence Distribution of Giant Planets*, submitted to AJ
- Jessica Ranshaw, Jennifer Burt, Xian-Yu Wang, et al. (incl. **Brandon T. Radzom**) 2026. *The Discovery of K2-232c: Divergent Formation Histories for Hot and Warm Jupiters Based on Outer Companion Eccentricity*, accepted to AJ, arXiv:2609.09077

Andrew Hotnisky, Rachel Fernandes, Kevin Hardegree-Ullman, et al. (incl. **Brandon T. Radzom**) 2026. *COUNTESS I: A Uniformly Vetted Catalog of Known and New Transiting Exoplanets in the TESS Northern Continuous Viewing Zone*, ApJS 286 42

Samuel Quinn, Joseph Rodriguez, Karen Collins, et al. (incl. **Brandon T. Radzom**) 2026. *TOI-2494 c and TOI-5143 c: a hot Saturn and a hot Jupiter with interior planetary companions*, AJ 171 359

R. Kent Honeycutt, Jeff Robertson, **Brandon T. Radzom** 2025. *Stunted Outbursts and Z Cam-like Behaviors in the Long-term Light Curves of Novalike Cataclysmic Variables*, ApJS 277 29

Cassandra Seltzer, Rudi Lien, **Brandon T. Radzom**, et al. 2025. *THUNDER: A Titan orbiter mission concept for the New Frontiers program*, PSJ 6 45

Jack Lubin, Xian-Yu Wang, Malena Rice, et al. (incl. **Brandon T. Radzom**), 2023. *TOI-1670 c, a 40 day Orbital Period Warm Jupiter in a Compact System, Is Well Aligned*, ApJL 959 L5

Xian-Yu Wang, Malena Rice, Songhu Wang, et al. (incl. **Brandon T. Radzom**) 2022. *The Aligned Orbit of WASP-148 b, the Only Known Hot Jupiter with a Nearby Warm Jupiter Companion, from NEID and HIRES*, ApJL 926 L8

NON-REFEREED

Brandon T. Radzom 2026. *Observational and Numerical Constraints on the Dynamical Origins of Hot Jupiters*, PhD Thesis, IU-Bloomington Dept. of Astronomy

Brandon T. Radzom 2020. *Characterizing AGN Activity in the SSA22 Field*, Senior Thesis, UW-Madison Dept. of Astronomy

Awards & Grants

ACCEPTED PROPOSALS

Co-I: NN-EXPLORE WIYN/NEID Semester 2026B (2 nights), NASA/NOIRLab. “RICHESE: Resolute Investigation to Characterize the Habitats of Exoplanets out into the Savanna Edges”

Co-PI: Indiana Space Grant Consortium 2025-2029 Cycle (\$5,550), NASA. “PyIU: Python Workshops to Forge Future Astronomers in Indiana”

Co-I: NN-EXPLORE WIYN/NEID Semester 2022A (8 nights), NASA/NOIRLab. “Probing Stellar Obliquities with NEID”

FELLOWSHIPS & HONORS

2026	Benjamin F. Peery Jr. Service Award , IU Astronomy	\$500
2026	Hollis & Grete Johnson Graduate Research Prize , IU Astronomy	\$500
2026	Visiting Graduate Student Research Fellowship , Caltech/IPAC	\$28,000
2025-2026	College of Arts and Sciences Dissertation Research Fellowship , IU	\$24,000
2025	Joseph & Frances Morgan Swain Graduate Fellowship , IU Astronomy	\$1,000
2024-2025	Sullivan Research Fellowship , IU Astronomy	\$31,000
2024	Frank and Margaret Edmondson Prize for Teaching , IU Astronomy	\$500
2023	Goethe Link Prize for Outreach and Public Education in Astronomy , IU	\$500
2022	College of Arts and Sciences Travel Award , IU	\$200
2020	Thesis of Distinction , UW College of L & S	
2020	Lowell Doherty Award for Excellence in Astronomy , UW Astronomy	\$500
2020	Member of the Dean's List , UW College of L & S	
2019	Member of the Dean's List , UW College of L & S	
2019	Liebenberg Family Undergraduate Research Scholarship , UW College of L & S	\$2,000
2019	David H. Durra Scholarship , UW College of L & S	\$3,000
2018	John Karl Scholz Sophomore General Scholarship , UW College of L & S	\$500

2017 **Member of the Dean's List**, UW College of L & S
2016 **Memorial Scholarship**, Anoka High School

\$ 500

Presentations

INVITED TALKS & COLLOQUIA

Hot and (Un)Bothered: Unveiling the Complex Dynamical Histories of Hot Jupiters. Astrophysics Colloquium, NASA/Jet Propulsion Laboratory (JPL), Dec. 4, 2025.

Hot and (Un)Bothered: Unveiling the Complex Dynamical Histories of Hot Jupiters. Astronomy Seminar, Department of Astronomy, Caltech, Dec. 1, 2025.

Hot and (Un)Bothered: Leveraging Compact Multi-planet Systems to Constrain the Origins of Hot Jupiters. Center for Exoplanets and Habitable Worlds Seminar, Department of Astronomy, Penn State University, Sep. 22, 2025.

Hot and (Un)Bothered: Leveraging Compact Multi-planet Systems to Constrain the Origins of Hot Jupiters. Astronomy Seminar, Earth and Planets Laboratory, Carnegie Institute, Sep. 12, 2025.

Measuring Stellar Obliquities To Constrain the Origins of Exoplanets. Virtual WIYN Board Meeting, Oct. 2023.

SEMINAR TALKS

COUNTESS: A Pipeline to Extract the Potential of the TESS Mission for Exoplanet Demographics. IPAC Lunch Seminar, Caltech/IPAC, Jun. 24, 2026.

Hot and (Un)Bothered: Unveiling the Complex Dynamical Histories of Hot Jupiters. Planetary and Exoplanetary Lunch Seminar, Department of Astronomy, University of Maryland, Nov. 17, 2025.

Hot and Cold: The Dynamical Origins of Hot Jupiters in the Era of TESS. TESS Science Talk, Kavli Institute for Astrophysics and Space Research, MIT, Nov. 12, 2025.

Hot and (Un)Bothered: Unveiling the Complex Dynamical Histories of Hot Jupiters. Exoplanet and Stars Seminar, Department of Astronomy, Yale University, Oct. 22, 2025.

Hot and (Un)Bothered: Leveraging Compact Multi-planet Systems to Constrain the Origins of Hot Jupiters. SEEC Exoplanet Seminar, NASA Goddard Spaceflight Center, Sep. 10, 2025.

Hot and (Un)Bothered: Leveraging Compact Multi-planet Systems to Constrain the Origins of Hot Jupiters. ESPF Seminar, Space Telescope Science Institute, Sep. 9, 2025.

NASA PSSS: A Lightning-Fast Lesson in Mission Design and the THUNDER That Follows. Astronomy Seminar, Astronomy Department, IU, Jan. 2024.

The X-ray Luminosity Function of Optically Narrow and Broad-line AGNs Out To $z \sim 4$. Astronomy Seminar, Astronomy Department, IU, Sep. 2020.

CONTRIBUTED TALKS

COUNTESS: A Pipeline to Extract the Potential of the TESS Mission for Exoplanet Demographics. 248th Meeting of the American Astronomical Society, Pasadena, CA, Jun. 14-18, 2026.

Scrutinizing Observational Signatures: New Constraints on the Dynamical Origins of Hot Jupiters. 247th Meeting of the American Astronomical Society, Phoenix, AZ, Jan. 3-8, 2026.

Hot or Not: Hot Jupiters' Isolation Is Not Unique to High-Eccentricity Tidal Migration. Great Lakes Exoplanet Area Meeting (GLEAM), UW-Madison, Nov. 6-7, 2025.

Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact Giant Systems. Carl Sagan Summer Workshop, Caltech, July 21-25 2025.

Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact Giant Systems. ERES IX, Cornell University, July 2024.

Post-formation Dynamics: A Mechanism to Explain the Companionship Properties of Hot and Warm Jupiters. GLEAM, IU, Oct. 27-28, 2023.

Post-disk Dynamical Evolution: A Mechanism to Explain the Companionship Dichotomy Between Hot Jupiters and Warm Jupiters. Division on Dynamical Astronomy Meeting #54, Michigan State University, May 2023.

PUBLIC TALKS

Exoplanets in the Media: Fact or Fantasy? Astronomy on Tap, Bloomington, IN, Sep. 2024.

Astronomy in the News: NASA's DART Mission. Astronomy on Tap, Bloomington, IN, Mar. 2024.

The Night Sky. Indiana Master Naturalist Astronomy Event, Bloomington, IN, Oct. 2021.

CONFERENCE POSTERS

Brandon T. Radzom, et al. (2026) "PyIU: A Python Workshop Series to Forge Future Astronomers in Indiana", 247th Meeting of the American Astronomical Society, Phoenix, AZ, Jan. 3-8, 2026. id.413.05

Brandon T. Radzom, et al. (2025) "To High-e or Not to High-e: Hot Jupiters in Compact Multi-Planet Systems Are Aligned", NOIRLab Solar System In Context. Sep. 29-Oct. 2, 2025.

Brandon T. Radzom, et al. (2025) "Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact Giant Systems", Carl Sagan Summer Workshop. July 21-25, 2025.

Brandon T. Radzom, et al. (2025) "PyIU Python Workshops: Forging Future Astronomers in Indiana", Spring 2025 Statewide Engagement Institute. Mar. 13, 2025. 10.5281/zenodo.15015534

Brandon T. Radzom, et al. (2024) "Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact TESS Systems", TESS Science Conference 3, July 29-Aug. 2, 2024. 10.5281/zenodo.13117605

Cassandra Seltzer, Rudi Lein, **Brandon T. Radzom**, et al. (2024) "THUNDER: A New Frontiers-class Titan orbiter mission concept from the NASA JPL Planetary Science Summer School", 55th Lunar and Planetary Science Conference, Mar. 11-15, 2024.

Brandon T. Radzom, Songhu Wang, and Bonan Pu (2022) "In Situ Origins of Hot Jupiters", Emerging Researchers in Exoplanet Science (ERES) VII, July 31-Aug. 2, 2022. 10.5281/zenodo.6944743

Brandon T. Radzom, Amy J. Barger, and Anthony J. Taylor. (2020) "Characterizing AGN Activity in the SSA22 Field", 236th Meeting of the American Astronomical Society, June 1-3, 2020. id.137.03

Minho Kwon, Christopher Young, Matthew Ebert, Sebastian Malewicz, **Brandon Radzom**, et al. (2018) "Progress toward entanglement of atomic ensemble qubits via Rydberg blockade", International Conference on Atomic Physics, July 22-27, 2018.

MEDIA INTERVIEWS

M. R. (2026), *Rare hot Jupiters could reveal how all giant planets form*, Eos, 107, <https://doi.org/10.1029/2026EO260070>. Published on 26 Feb. 2026

Graduate student's summer experience at NASA's Jet Propulsion Laboratory culminates in mission presentation to scientists, leaders, College of Arts + Sciences News, news.iu.edu/college/live/news/35938-graduate-students-summer-experience-at-nasas-jet<https://doi.org/10.1029/2026EO260070>. Published on 15 Sep. 2023

Teaching Experience

Instructor of Record

IU Astronomy

- Astronomy 100: The Solar System (Summer 2023 & 2024)

Associate Instructor

IU Astronomy

- Astronomy 451: Stellar Astrophysics (Spring 2024)
- Astronomy 305: Modern Observational Techniques (Fall 2023)
- Astronomy 107: The Art of Astronomy (Spring 2023, Fall 2020)
- Astronomy 100: The Solar System (Spring 2022)
- Astronomy 103: Search For Life In The Universe (Fall 2021)
- Astronomy 222: General Astronomy II (Spring 2021)

Guest Lecturer

IU Astronomy

- Astronomy 222: General Astronomy II (Spring 2025)
- Astronomy 100: The Solar System (Spring 2024, Spring 2023)
- Astronomy 103: Search for Life in the Universe (Spring 2024)
- Astronomy 515: Exoplanets and Orbital Dynamics (Fall 2023)

Community Involvement & Leadership

SERVICE

2025-2026	Reviewer , AAS + A&A Journals	
2022-2026	Founder and Lead Organizer , PyIU: A Workshop Series on Python Essentials	Bloomington, IN
2025	Hands-on Session Helper , Carl Sagan Summer Workshop	Pasadena, CA
2025	Steering Committee Member , Concerned Scientists at IU	Bloomington, IN
2025	AAS Advocacy Representative , AAS Congressional Visits Day	Washington, D.C.
2024-2025	Pre-Conference Survey Lead , ERES Mission & Vision Committee	
2023-2025	Mentor , Alice-Palma Undergraduate Research Program	Bloomington, IN
2023-2025	Seminar Talk Coordinator , IU Astronomy Graduate Leadership	Bloomington, IN
2022-2025	Undergraduate Research Coordinator , IU Astronomy Graduate Leadership	Bloomington, IN
2023	SOC & LOC Member, Session Chair , GLEAM 2023 Conference	Bloomington, IN
2019-2020	Vice President , UW-Madison Astronomy Club	Madison, WI
2018-2019	Volunteer Coordinator , UW-Madison Astronomy Club	Madison, WI

PUBLIC OUTREACH

2022-2026	PyIU: A Workshop Series on Python Essentials , Instructor, code developer	Bloomington, IN
2021-2026	IU Science Fest , Astro demo leader	Bloomington, IN
2025	Kirkwood Observatory Community & School Partnerships Tour , Tour guide	Bloomington, IN
2025	Kirkwood Observatory Class & Staff Tours , Tour guide	Bloomington, IN
2021-2025	Kirkwood Observatory Open Nights , Telescope operator & tour guide	Bloomington, IN
2024	Astronomy on Tap , Public speaker: <i>Exoplanets in the Media</i>	Bloomington, IN
2024	Boy Scout Astronomy Night , Observatory tour Guide	Bloomington, IN
2024	Astronomy on Tap , Public speaker: <i>NASA's DART Mission</i>	Bloomington, IN
2024	Eclipse IU , Solar eclipse viewing aid and guide	Bloomington, IN
2023	Kirkwood Observatory High School Class Tour , Telescope operator & tour guide	Bloomington, IN
2023	STEAM Night at McCormick's Creek Elementary , Astro demo leader	Spencer, IN
2023	International Day of Women and Girls in Science , Astro demo leader	Bloomington, IN
2022, 2023	Indianapolis Children's Museum STEM Exploration Day , Astro demo leader	Indianapolis, IN
2022	Boys and Girls Club: Streets Paved With Gold (Alpha Phi Alpha) , Astro demo leader	Bloomington, IN
2021	Indiana Master Naturalist Event , Guest speaker, telescope operator	Bloomington, IN
2020	Girl Scout Astronomy Night , UW-Madison Astronomy Club volunteer	Madison, WI
2019	Moon Over Monona Terrace , UW-Madison Astronomy Club volunteer	Madison, WI
2019	My UW Days , UW-Madison Astronomy & Physics Dept. representative	Madison, WI
2019	UW Space Place OAO-2 Anniversary Event , Observatory historical guide	Madison, WI

2018 **UW STEM Immersion Day**, UW Astronomy/ Astronomy Club representative

Madison, WI

2017 **UW Space Place Family Science Night**, Physics and Astro demo leader

Madison, WI

PROFESSIONAL DEVELOPMENT

Fall 2022-Summer 2026: PyIU: A Workshop Series on Python Essentials, *IU Astronomy Dept.* Founder and lead organizer of the PyIU program: a free, graduate student-led Python workshop series for undergraduate and high school students (serving over 200 students to date) intended to build confidence in pursuing STEM and equip with skills to engage in astronomy and physics research. This program was also awarded funding by NASA's Indiana Space Grant (co-PIs: myself and Samir Salim) for the 2025-2026 academic year. All content is posted to a public GitHub repository: github.com/BrandonRadzom/pyiu.

Summer 2025: Carl Sagan Summer Workshop Silver Jubilee: Exoplanet Demographics, *NExSci/Caltech* Participant, POP poster presenter, and helper for the hands-on "Exoplanet Occurrence Rates" and "Distant Giant Planets with Astrometry" sessions. Workshop link: <https://nexsci.caltech.edu/workshop/2025/>.

Summer 2023: JPL Planetary Science Summer School, *JPL*. Received training on mission formulation, and worked with a large, multidisciplinary team and consulted with JPL's Team-X to design a concept for a New Frontiers-class orbiter at Saturn's moon, Titan. Assumed the role of Deputy PI, Science Chair, and Science Objective Lead. Presented the concept to a review panel at JPL (published also in the *Planetary Science Journal*). Program link: <https://www.jpl.nasa.gov/edu/internships/apply/nasa-science-mission-design-schools/>

Summer 2023: Code/Astro Software Engineering Workshop, *Northwestern University*. Trained on git and GitHub, de-bugging tools, releasing code to PyPI and GitHub, documenting code (using docstrings & Sphinx), and software testing. Co-developed a pip-installable package for planning observations at IU's Kirkwood Observatory called `kirkwoodnight`. Workshop link: semaphore.github.io/codeastro.